

DEPARTMENT OF MINING ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY (ISM) DHANBAD

LECTURE PLAN

Class : IV SEMESTER - B. TECH. (ME)

Session : 2020-21

Subject : **ROCK MECHANICS** (MNC-205)

L-T-P : 3 – 1 - 0

MAXIMUM MARKS: 100

MARKS ALLOCATION: Mid Semester Exam: 30
 Assignment: 5

End Semester Exam: 45
 Project: 12
 Quizzes/Class test: 8

Unit No.	Details of the unit	No. of Lectures
Unit 1	Geotechnical Investigations and Classification of rock mass and soil • Engineering properties of intact rock – physico-mechanical • Overview of requirements, methods and analysis of engineering-geological investigation methods • Rock mass classification methods and their applications • Soil classification methods and their applications.	8
Unit 2	Concept of Stress, strain and failure of rock • Analysis of stress, Analysis of strain, Constitutive relations, Parameters influencing strength/stress-strain behavior • Failure Criteria for Rock and Rock Mass Classical theories of rock failure: Coulomb's criterion, Mohr's criterion, Griffith's theory, Empirical failure criteria. Behaviour of jointed rock mass. • Pre-mining state of Stress Stresses in rock mass, Factors influencing the in situ state of stress, Estimating in situ stresses; Methods of Stress determination- Hydro fracturing, stress relief methods. • Overview of shear strength and compressibility of soil	14
Unit 3	Rock mass properties • Strength and Deformability of Rock Mass In situ shear tests; Evaluation of shear strength; In situ bearing strength test; In situ deformability tests- Plate Loading Test, Plate Jacking Test and Borehole Jack Tests • Field measurement of soil properties	2
Unit 4	Response of rock mass and soil to excavation • Response of rock mass to Excavations Underground, Induced stresses and displacements around single opening in rock mass; • Design of excavations in massive elastic rock • Design of mine pillars • Ground support interaction analysis and reinforcement of ground (rock mass and soil), selection and design of support systems. • Slope Engineering: Slope failure and causes; Basic approaches to slope stability analysis and stabilisation • Monitoring of Excavation Stability: Purpose and nature of monitoring, Instrumentation and monitoring systems - Load; Stress and Deformation measuring devices; Interpretation of monitoring data; Practical aspects of monitoring.	7
Unit 5	Groundwater flow: Permeability and pressure Groundwater flow within soil and rock masses; Permeability conditions; Influence of groundwater soil and rock mass behaviour; Measurement of groundwater pressure and permeability	2
Unit 6	Ground improvement; grouting, fore polling, pre-reinforcement, shotcreteing and other	2
Unit 7	Subsidence: Types, causes and impacts of subsidence; Factors influencing subsidence; subsidence prediction and control	2
Unit 8	Numerical methods : Basics of FEM, FD and BE methods	2
	Total theory classes	39
	Tutorials	13

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